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FOUNDATION MODEL-BASED MULTIMODAL AI FOR PROSTATE CANCER RISK PREDICTION

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Abstract

***Purpose:** Accurate diagnosis and risk stratification of prostate diseases remain challenging due to the heterogeneous nature of multi-center data and the limited utilization of multimodal clinical information. This study aims to develop a unified artificial intelligence framework that integrates imaging and clinical data for comprehensive prostate disease analysis and clinical decision support.

***Methods and Materials:** A multi-center cohort of ~6,000 prostate MRI cases with clinical data was collected from institutional and public datasets. Preprocessing included registration, normalization, prostate localization, and ROI expansion. A foundation model was pretrained using MAE and BYOL. Imaging features were fused with ClinicalBERT-encoded clinical data via cross-attention. A multi-task framework addressed classification, invasion, and functional prediction. A LangChain-based AI agent supported clinical decision-making.

***Results:** The proposed multimodal model achieved the best performance in distinguishing benign prostatic hyperplasia (BPH) from prostate cancer (PCa), with an AUC of 0.854 ± 0.011 , outperforming both the imaging-only model (T2-weighted MRI: AUC = 0.848 ± 0.010) and the clinical-only model (AUC = 0.81 ± 0.02). These results demonstrate that multimodal fusion provides complementary information beyond imaging features alone. The framework also showed stable performance across multiple tasks, indicating good generaliz

ability.

***Conclusions:** This study presents a comprehensive multimodal AI framework that combines foundation model pretraining, cross-modal fusion, and agent-based clinical integration for prostate disease analysis. The proposed approach improves diagnostic performance and provides a scalable pathway toward intelligent clinical decision support systems in prostate imaging.

***Clinical Relevance/Application (NOT REQUIRED for Quality Improvement Reports, enter "N/A" into this section):** The multimodal AI framework improves non-invasive differentiation between benign prostatic hyperplasia and prostate cancer. By integrating MRI and clinical data, it enhances risk assessment, potentially reducing unnecessary biopsies and supporting earlier detection. Multi-task prediction enables evaluation of tumor aggressiveness and invasion risk, improving clinical decision-making and workflow efficiency.

Category	Complete Genitourinary Imaging -> GUPAIR - Prostate: AI/Radiomics
Questions	Complete Trainee Research Prize Medical Student Disclosure of "Off-Label" usage No, I do not intend to discuss off-label uses IRB / IACUC Response Human subject under a waiver from the IRB Has this work been previously presented or published? No 2nd Format Opportunity: Yes, I would be interested if accepted to showcase my work in an additional format (2 meter Hardcopy Backboard).

Attached Files

summary (PDF, 4117586 bytes)